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SUMMER SEMESTER-B.TECH (2026)
END SEMESTER EXAMINATION

Course Code: CMMTC04

Course Title: Engineering Mathematics

Time: 3 Hours

Max. Marks: 50

Note: Attempt all questions. Attempt any two parts from each question. Missing data (if any) may be suitably assumed.

Q.No	Attempt all the five questions.	Marks	CO																
1(a)	Solve the simultaneous equations by Laplace Transform: $(D^2 - 3)x - 4y = 0$ and $x + (D^2 + 1)y = 0$ for $t > 0$, given that $x = y = \frac{dy}{dt} = 0$ and $\frac{dx}{dt} = 2$ at $t = 0$.	5	1																
1(b)	Find the Laplace transform of the function: $f(t) = \frac{\cos \sqrt{t}}{\sqrt{t}}$.	5	1																
1(c)	State Convolution theorem. Evaluate the inverse Laplace transform : $L^{-1} \left(\frac{s}{(s^2 + a^2)^2} \right)$.	5	1																
2(a)	Express $f(x) = \cos(\omega x)$ for $-\pi < x < \pi$ as a Fourier series, where ω is a fraction.	5	2																
2(b)	Let $f(x) = \begin{cases} \omega x, & 0 < x \leq \frac{l}{2} \\ \omega(l - x), & \frac{l}{2} \leq x < l \end{cases}$. Show that $f(x) = \frac{4\omega l}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)^2} \sin \frac{(2n+1)\pi x}{l}$.	5	2																
2(c)	Determine the Fourier series up to the second harmonic for the following data: <table border="1"><tr><td>x</td><td>0°</td><td>30°</td><td>60°</td><td>90°</td><td>120°</td><td>150°</td><td>180°</td></tr><tr><td>$f(x)$</td><td>0</td><td>5224</td><td>8097</td><td>7850</td><td>5499</td><td>2626</td><td>0</td></tr></table>	x	0°	30°	60°	90°	120°	150°	180°	$f(x)$	0	5224	8097	7850	5499	2626	0	5	2
x	0°	30°	60°	90°	120°	150°	180°												
$f(x)$	0	5224	8097	7850	5499	2626	0												
3(a)	Find the solution of the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the following boundary conditions: (i) $u \rightarrow 0$ as $y \rightarrow \infty$ for all x , (ii) $u = 0$ at $x = 0$ for all y , (iii) $u = 0$ at $x = l$ for all y , (iv) $u = x(l - x)$ if $y = 0$ for all $x \in (0, l)$.	5	3																
3(b)	Solve the heat equation $\frac{\partial u}{\partial t} = k^2 \frac{\partial^2 u}{\partial x^2}$, if $u(0, t) = u(l, t) = 0$ and $u(x, 0) = 50$.	5	3																

Q.No	Attempt All the five questions.	Marks	CO
3(c)	A string is attached and fastened to two points l distance apart. Motion is started by replacing the string in the form $u = a \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Find the displacement $u(x, t)$ from one end.	5	3
4(a)	Using Cauchy Residue theorem, evaluate $\int_C \frac{12z-7}{(z-1)^2(2z+3)} dz$, where $C : z = 2$.	5	4
4(b)	Find the analytic function $f(z) = u + iv$ if $u - v = \frac{\cos x + \sin x - e^{-y}}{2 \cos x - 2 \cosh y}$, find $f(z)$, subject to condition $f(\frac{\pi}{2}) = 0$.	5	4
4(c)	Using Cauchy Integral formula, evaluate the values of $f(3)$, $f'(1-i)$ and $f''(1-i)$, where $f(s) = \oint_C \frac{3z^2+7z+1}{z-s} dz$ where C is the circle $x^2 + y^2 = 4$.	5	4
5(a)	Evaluate $\iint_S (\text{curl } \vec{F}) \cdot \hat{n} dS$ where $\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy -plane.	5	5
5(b)	Show that the vector function $\vec{v} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$ satisfies $\nabla \cdot \vec{v} = \frac{2}{\sqrt{x^2 + y^2 + z^2}}$ and $\nabla \times \vec{v} = 0$.	5	5
5(c)	Evaluate $\iint_S \vec{A} \cdot \hat{n} dS$ where $\vec{A} = z\hat{i} + x\hat{j} - 3y^2z\hat{k}$ and S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.	5	5